

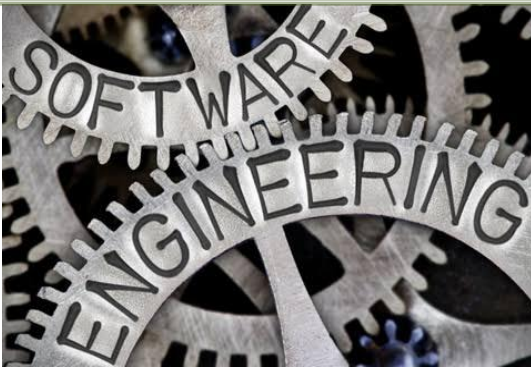


NILE UNIVERSITY OF NIGERIA

HONORIS UNITED UNIVERSITIES

2025-2026

STUDENT HANDBOOK



Software Engineering



Software Development process



Programming Languages



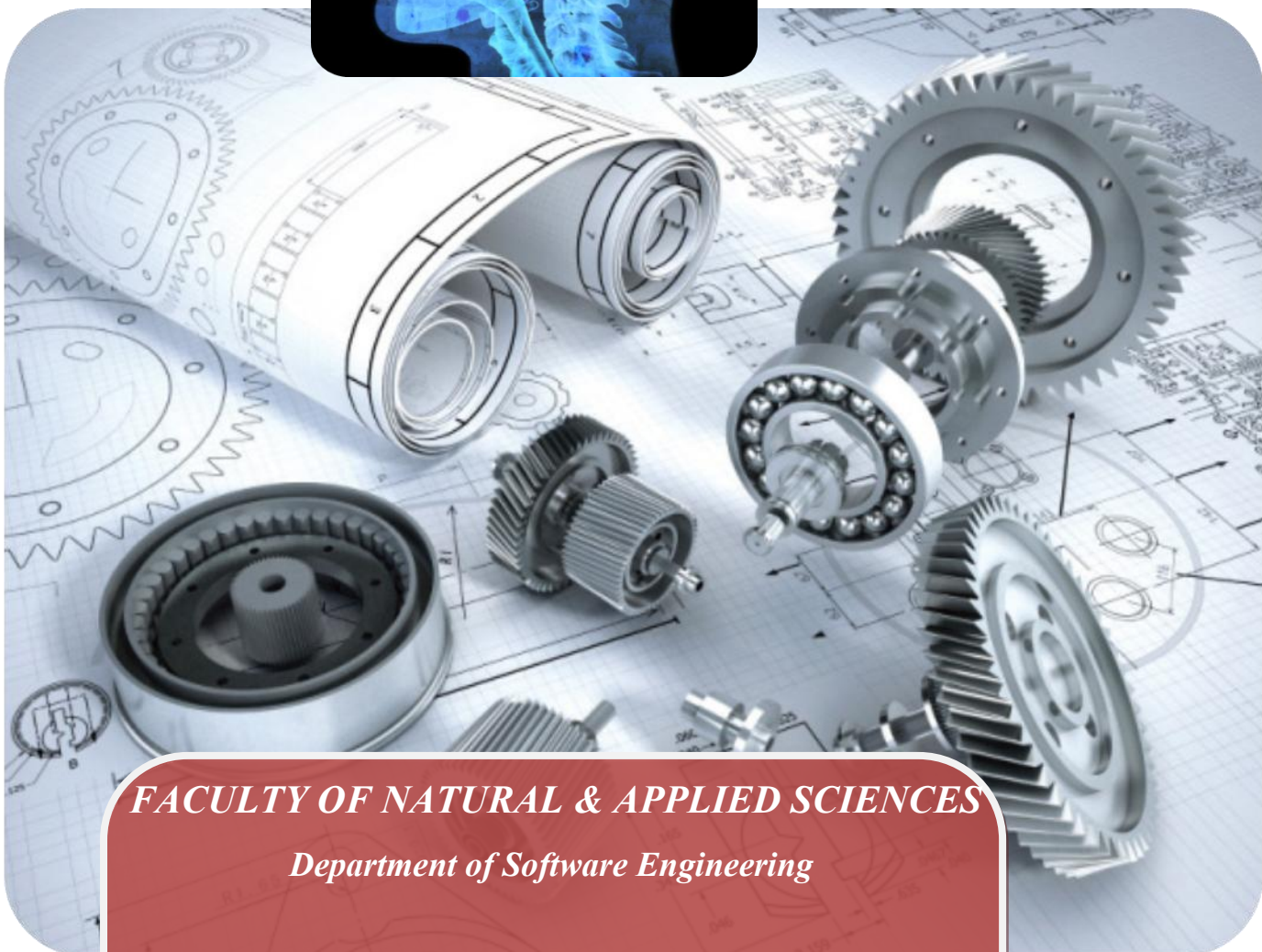
Application Development



Software Engineering
“Gives life to creative expressions.”



Nile University of Nigeria



FACULTY OF NATURAL & APPLIED SCIENCES

Department of Software Engineering

STUDENT HANDBOOK

2021-2022

ABUJA

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Nile University of Nigeria

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2021/2022

ABUJA

Ignorance of the rules and regulations of the university shall not be a defence to any erring student. All students are therefore required to familiarise themselves with the rules and regulations as contained in this departmental student handbook that accompanies

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MESSAGE FROM THE HEAD OF DEPARTMENT

On behalf of the Department of Software Engineering (SE) and Nile University of Nigeria (NUN), Abuja, Nigeria, I extend a warm welcome to all the students of the Department of Software Engineering. Ours is a department of deep thinkers. I hope that your experience with the Department and the Faculty will be a very rewarding one. By enrolling in the Software Engineering Program, you are set to embark on a journey into the exciting world of Science, Technology and Innovation and specifically architecting software applications for various systems.

The mission of the SE Department for undergraduate education is to provide the student with the fundamental theory, experimentation and the engineering that form the basis for the design, development and testing of software systems. The Software Engineering Curriculum has been carefully designed to give the student both depth and breadth of learning while satisfying both local and international standards.

The duration of the Program is normally 4 years leading to the award of Bachelor of Science (B.Sc.) degree in Software Engineering. The Program relies on a hands-on practical-oriented curriculum to train software Engineers and Professionals that will be fully confident of what they learned over the 4 years of study at NUN.

You will learn entrepreneurial skills that will make you a potential job creator rather than a job seeker when you graduate. The University is endowed with internationally renowned academics in various fields of science and technology, majority of whom are always ready to attend to your academic challenges and guide you towards the full development of your potentials. Our programs have been uniquely designed to focus on character-building, personality-development and critical-thinking, apart from a rigorous academic and research curriculum.

The learning environment is rich in facilities, techniques, technologies and tools that ensure easy assimilation, enhance feedback, and interaction. This encourages team work and develops the student's collaborative skills.

This Handbook is designed to provide you with specific information about the Faculty, general information about services and facilities, and a quick reference on the academic rules and regulations that govern your studies in the Department of Software Engineering at Nile University of Nigeria. Other detailed information about your programme will be posted on the departmental website and on the Departmental Notice Board. Please, refer to the Handbook as the information it contains will prove useful throughout the duration of your study.

On behalf of the Department and the Faculty, I wish you a pleasant educational career at Nile University of Nigeria, Abuja.

Salihu Ibrahim Anka, Ph.D.
HOD, Software Engineering

PURPOSE OF HANDBOOK

This Handbook has been developed for the purpose of informing you of the policies, procedures, regulations for the Software Engineering program. This is to ensure that all students receive accurate, clear, useful and timely information about the available programs of study, learning facilities and related information. We hope that it will assist you in making your learning experience pleasant and productive. It is the responsibility of each student and parent/sponsor to read, understand, and abide by this Handbook.

All policies listed in this Handbook are subject to change by the Management of the NUN or through administrative action. Students will be notified when such changes occur. The changes will also be posted on the department or University home page - www.nileuniversity.edu.ng, and at strategic locations to be identified. More information on the Policies referenced in this Handbook can always be accessed on the Institute website - www.nileuniversity.edu.com.

Nile University of Nigeria, Abuja, is committed to acting in accordance with Equal Opportunities, Equality and Diversity Policies of the government. It is intended that all students should receive equal treatment irrespective of race, colour, religion, gender identity, age, nationality, ethnic origin, political beliefs, marital or parental status, disability, physical, mental, emotional, social or family background.

Disclaimer

The information provided in this Handbook is correct at the time of printing [October, 2021]. The Department reserves the right to make amendments to:

- a) the contents of the Handbook and in particular to the timetable, location and methods of delivery of the content, syllabus and assessment of any of its programs, module specifications of this Handbook, and/or on the department's website; and
- b) its rules, regulations, policies, procedures and fee structures;

provided that such amendments are, (i) as a result of student demand (or lack thereof), (ii) as a result of unforeseen events or circumstances beyond the department's control or, (iii) are deemed reasonably necessary by the University Management.

In the event that amendments are made, the Department shall take reasonable steps to notify the student as soon as is reasonably possible.

List of Academic Staff and their Qualifications

Professor/Reader

N ₀	Name	Specialization	Discipline	Qualification	Rank
1	Joshua Abah	Cyber Security	Computer Science	B.Sc./MSc./Ph.D.	Professor
2	Saleh Abdullahi	Operating Systems	Computer Science	B.Sc./MSc./Ph.D.	Professor
3	Nwojo Agwu Nnanna	Numerical Analysis/Analysis of Algorithms	Mathematics	B.Sc./MSc./Ph.D.	Professor

Senior Lecturer

N ₀	Name	Specialization	Discipline	Qualification	Rank
1	Salihu Ibrahim Anka	Software Quality and Testing	Software Engineering	B.Sc./MSc./Ph.D.	HOD/Reader
2	Bilkisu Muhammad Bello	Database Systems/Cloud	Computer Science	B.Sc./M.Sc./Ph.D	Reader
3	Muhammad Aliyu Suleiman	Machine Learning	Computer Science	B.Sc./M.Sc./Ph.D.	Reader

Lecturer 1 & Below

N ₀	Name	Specialization	Discipline	Qualification	Rank
1	Asmau Usman	Software Testing	Software Engineering	B.Sc./M.Sc./Ph.D.	Lecturer I
2	Emmanuel Aderemi	Machine Learning	Computer Science	B.Sc./M.Sc./Ph.D.	Lecturer II
3	Shuaibu Rabi Aliyu	Cloud Computing	Computer Science	B.Sc./M.Sc.	Lecturer II
4	Hauwa Ibrahim Aminu	IoT	Information System	B.Sc./M.Sc.	Asst. Lecturer
5	Austine Ogar	Artificial Intelligence	Computer Science	B.Sc./M.Sc.	Asst. Lecturer
6	Abdulsalam Nur	Software Engineering	Computer Science	B.Sc./M.Sc.	Asst. Lecturer

1 VISION, MISSION, PHILOSOPHY, OBJECTIVES AND STUDENT OUTCOMES

1.1. A Brief History of the Department

The Department of Software Engineering (SE) at Nile University of Nigeria, Abuja was established in 2018. At the undergraduate level, it currently offers program leading to the Bachelor of Science (B.S.) in Software Engineering. The program prepares students equally well for graduate work or professional positions in the computing and information technology fields and provides a concentrated approach to Software Engineering, while encourages students to combine Software Engineering with studies in another field. The SE program is oriented towards students with interest in applying computing concepts and information technology theories to practical applications in required fields.

Since Software Engineering applies to so many facets of modern life, students in the program are encouraged to utilize their elective courses to pursue interests in other areas of the fields. So the SE Department also may offer joint courses or degree programs with various other departments.

1.1.1. What is Software Engineering?

Software Engineering is the systematic application of engineering approaches to the development of software. It is the scientific and practical approach to computation and its applications and the systematic study of the feasibility, structure, expression, and mechanization of the methodical procedures (or algorithms) that underlie the acquisition, representation, processing, storage, communication of, and access to information. An alternate, more succinct definition of Software Engineering is the study of automating algorithmic processes that scale.

Software engineering experts are in demand and will likely continue to be in demand as the technology-focused business environment expands. Software engineers are the creative minds that analyse the needs of computer users to design and develop software applications to answer these needs. They have extensive knowledge of programming languages, software development, and computer operating systems, and they apply engineering principles to software creation. By applying these engineering principles to every stage of the development process, from requirements analysis to the software process, they can create customized systems for individual clients. Just as a civil engineer will make sure that a bridge has a solid foundation, a software

engineer will also begin with a thorough study of requirements and work through the development process in a systematic way.

The fundamental skills that are necessary for software engineers are familiarity with modern software architecture and patterns, programming in a variety of languages such as Python, JavaScript, Java, C, C++, C#, Ruby, familiarity with programming frameworks, familiarity with database technologies such as SQL, Postgres, MongoDB, and strong critical thinking abilities and problem-solving skills. Software Engineers also design and analyse algorithms to solve a variety of human problems and design applications that perform well on different mobile devices, that are easy to use, and that uphold security measures.

1.2. Vision

The Department envisions to be a competitive entity in teaching and research in software engineering in the country and the region and to produce graduates with international recognition for their knowledge and skills, capable of taking leadership positions in the field of software engineering.

1.3. Mission

The mission of the Department is to keep the academic and research programs on a par with internationally recognized programs in the discipline incorporating the rapid changes taking place in the industry, train the students in the design, construction, implementation, and maintenance of software of modern computing systems, specialized business applications and computer-controlled equipment that constitute the discipline of software engineering. The department intends to achieve these by making sure that the graduates:

- i. Understand the basic principles and practices in developing and maintaining software systems that behave reliably and efficiently and satisfy all the requirements that customers have defined for them;
- ii. are capable of creatively applying their scientific skills and engineering principles to solve problems typical in the career of software engineering;
- iii. are competent in communicating their ideas clearly and succinctly in all forms; and

- iv. are motivated to engage in lifelong learning which will be essential for progressing in their career.

1.4. Philosophy

The general philosophy is to produce graduates with high academic standard and adequate practical background for self-employment as well as being of immediate value to the industry and the community in general. The Department relies on a balanced hands-on practical-oriented software engineering curriculum aimed at producing graduates with heightened entrepreneurial skills who become job creators than job seekers. The department intends to use the best available materials in the industry and academia to expose the students to the rapidly changing computing industry

1.5. Objectives

The objectives of the Software Engineering program conform to the goals and objectives of the National Policy on Industrialization and Self-Reliance and are geared towards:

- (i) The development of a thorough practice in Software Engineering training;
- (ii) Provision of broad-based training in Software Engineering at the early stages of the program
- (iii) Adequate training in human and organizational behavior and management;
- (iv) Introduction to Entrepreneurial education and training;
- (v) Close association of the program with industry in the country.

1.6. Student Learning Goals

- Fundamentals - The student majoring in Software Engineering must achieve an understanding of the nature and organization of numerical and symbolic

computation along with proficiency in the specification and creation of complex computer programs.

- **Theory** - The student should also become well-grounded in the theoretical concepts that underlie current and potential future techniques and applications. Requirements supporting this goal are core courses and specific courses in mathematics, statistics, and operations research.
- **Advanced Work** - Students attain more advanced knowledge of Software Engineering and related fields in their particular areas of interest, by specified senior-level Software Engineering courses and additional courses from a list in Software Engineering, engineering, mathematics, and philosophy.
- **Integration** - The student is expected to learn to assist people in other professions to address their concerns by the design, implementation, and maintenance of computational systems. To this end, students take specific courses in ethics, communication, composition, and software engineering, in addition to carrying out writing-across-the-curriculum assignments and fulfilling the University breadth requirement.

1.7. Learning Outcomes

- a. Software development skills.** Graduates of the Software engineering program will be familiar with modern software architecture and patterns, programming frameworks and be familiar with programming in popular languages such as Python, JavaScript, Java, C, C++, C#, Ruby.
- b. Programming skills.** Graduates will be able to write code, design programs, and implement processes to create products and solutions for businesses and government entities.
- c. Mathematics for Development of Computing Systems.** Graduates will demonstrate the ability to apply knowledge of mathematics to develop and analyse computing systems.

- d. **Data Management.** Graduates **will** demonstrate the ability to use database technologies such as SQL, Postgres, and MongoDB to store and manage large amount of data in software systems.
- e. **Problem specification.** Graduates will have the conceptual knowledge and background to be able to analyse a problem and identify and define the computing requirements for its solution.
- f. **Solution implementation.** Graduates will have the ability to implement a computer-based system, process, component or program.
- g. **Teamwork.** Graduates will be able to work effectively in teams in designing and implementing software systems.
- h. **Ethical issues.** Graduates will be aware of key ethical issues affecting Software Engineering and their responsibilities as Software Engineering professionals.
- i. **Oral communication.** Graduates will demonstrate the ability to orally communicate ideas and concepts clearly and in an organized manner.
- j. **Written communication.** Graduates will demonstrate the ability to write clear system documentation, user documentation and research reports.
- k. **Social impact.** Graduates will have an understanding of the impact of computers in society.
- l. **Theory and Concepts.** Graduates will have a solid understanding of the concepts underlying Software Engineering.
- m. **Technical skills.** Graduates will demonstrate the ability to use current techniques, skills and tools for computing practice.

1.8. Career Opportunities

Software Engineering Jobs

With the emergence of computing in every sector of life, graduates entering the workforce with a degree in Software Engineering will find many career opportunities in business, industry, government, and academia, and particularly in organizations with a science emphasis. These opportunities include traditional and emerging careers such as

application programming, systems programming, systems analysis, systems administration, bioinformatics, network administration, and computer modelling.

A well-planned set of science and supporting electives can also position a student for graduate study in a computing sub-discipline of a natural or applied science or mathematics including bio-informatics, computational chemistry or physics, or scientific visualization.

Internships and Co-ops

Students in the Software Engineering department programs can incorporate a cooperative educational or internship experience into their academic program. These professional, career-related experiences in an organizational setting are valuable for a student's professional career development, and students may receive academic credit that applies to their degree program.

1.9. Colour of Faculty Hoods

Green

2. ADMISSION REQUIREMENT

There are two sets of diversified student types that the university is targeting - the Nigerian students and the international students. The two groups have different criterion of admission.

2.1. Nigerian Students

All Nigerian admissions into the University shall be through the Joint Admissions and Matriculation Board (JAMB). Admissions shall be open to all irrespective of nationality, sex,

physical challenge, religion and race. A letter of recommendation from the previous Senior Secondary School Counsellor/ Principal is also required.

The University shall conduct its own entrance exam if necessary. This Post JAMB screening will involve written test and interview to assess the student's suitability or otherwise for admission and the candidate shall pay the specific amount only for the same. The results of these tests shall be declared within 15 days of the Exam date.

The entry requirements shall be at least credit level passes in five subjects including English Language, Mathematics, and Physics to form the core subjects with credit in any other two relevant science subjects, at the Senior Secondary School Certificate or its equivalent. In addition, an acceptable pass in the University.

Candidates with two A level passes (graded A-E) at the Advanced Level in one or more relevant subjects (Chemistry, Mathematics and Physics) may undertake the three – year degree programme into 200-level.

Additionally, admission into an undergraduate degree programme of the University shall be based on the Faculty and Departmental requirements stated below.

Table 1: Admission Requirements

Programme	Admission Requirements		Remarks
	U.M.E	Direct Entry	
Software Engineering	Five credit passes at SSCE (or its equivalent) including English, Mathematics, Physics.	• “A” level passes in Mathematics, Physics. • OND/NCE in Mathematics and Physics.	UTME subjects are Mathematics, Physics and any two of English, Biology, Chemistry

2.2. Foreign Students

The university proposes to admit foreign students through the following:

1. Scholastic Aptitude Test (SAT) – conducted in over 130 countries as an international entrance exam for admission into Universities the United States and American University abroad),
2. OSS - Student Selection and Placement System- The university entrance exam conducted in Turkey that is equivalent to JAMB - Nigeria),
3. YOS (another internationally recognized entrance exam conducted across 25 European Countries for admissions across Australia and Europe), or
4. Any other internationally recognized University Entrance Exam, equivalent of the JAMB in Nigeria.

In order to comply with the admission policy into Nigerian Universities, all foreign students admitted into NUN will be subsequently cleared with JAMB.

2.3. Direct Entry

Direct Entry candidates are those who have completed one of the following: OND, HSE, HND, ‘A’ level GCE, SBS, and IJMB or any other higher qualification. For this method of entry to NUN, candidates who have 9 points or above will be admitted to 200-Level based on the three subjects taken. Generally, those candidates who have below 9 points will be admitted to commence at 100-Level.

2.4. Inter/Intra-faculty Transfer

A student in 200 or 300 level can change department within/outside the faculty. This can be done by completing the “CHANGE OF DEPARTMENT FORM”. The time to change department is during the registration period in first semester of the student’s 200 or 300 level.

3. ACADEMIC RULES AND REGULATIONS

3.1. Duration of Programmes

Undergraduate degree programmes of the University shall have varying durations based on guidelines of National Universities Commission (NUC) Minimum Academic Standards. The durations shall be as follows:

- a) A minimum of eight semesters and a maximum of twelve semesters for a four year degree programme in the Faculty of Natural and Applied Sciences.
- b) A student accepted on transfer shall be obligated to spend at least two sessions in the university before graduating; and
- c) Any period of authorized withdrawal shall not be included in the normal duration of an academic programme.

3.2. Semester System and Course Credit System

The University runs the semester system for its academic programmes. Each academic session comprises two semesters and each semester lasts for a period of eighteen (18) weeks consisting of one (1) week for registration, fifteen (15) weeks for lectures and two (2) weeks for examinations. One week of lectures shall consist of 40 hours of classroom and/or laboratory activities. In line with the guidelines of the NUC, the University also operates the course credit system. One credit unit is defined as one hour of lecture or tutorial per week per semester and its equivalents shall be: two hours of seminars; three hours of laboratory or field work; clinical practice/practicum or studio practice; six hours of teaching practice; and one week industrial attachment.

The University adopts a course numbering system that uses a prefixed three-letter code identifying the department offering the course followed by a three-digit number identifying the course. Odd-ended and Even-ended numbers are used to identify first semester and second semester courses, respectively.

The University's Courses are numbered as follows:

Table 2: Undergraduate Course Levels

1	100 Level:	101 – 199	1 st Year
2	200 Level	201 – 299	2 nd Year
3	300 Level:	301 – 399	3 rd Year
4	400 Level:	401 – 499	4 th Year

3.3. Techniques of Student Assessment

- a) **Practical:** By the nature of the disciplines in Natural and Applied Sciences, laboratory practical are very important in training of students. To reflect this importance of practical work, a minimum of 9 hours per week (3 credits) should be spent on students' laboratory practical. Furthermore, it is very important to determine performance of the student in the practical component of the programme. To achieve this, all the laboratory practical has been lumped together to form a course which the student must pass. It is expected that the weighting given in the various courses is reflected in number and nature in the design of the experiments. These practical must follow the trend in the current development of the programmes.
- b) **Tutorials:** There should be one hour of tutorial for every four hours of lecture. Thus a course of one credit unit should comprise 12 hours of lecture and three hours of tutorials.
- c) **Continuous Assessments:** It shall be done through essays, quizzes, tests and practical exercises.
- Scores from continuous assessment shall normally constitute 30 per cent of the final marks for courses which are primarily theoretical.
 - For courses which are partly practical and partly theoretical, scores from continuous assessment shall constitute 50% of the final marks.
 - For courses that are entirely practical, continuous assessment shall be based on a student's practical work or reports and shall constitute 100 percent of the final marks.

d) **Examinations:** In addition to continuous assessment, final examinations are normally given for every course at the end of each semester. The final grading system may be based on the following breakdown:

- Final Examination: 60% - 70%
- Continuous assessment (Quizzes, Tutorials, Homework, Tests): 30% - 40%
- Each course shall normally be completed and examined at the end of the semester in which it is offered.
- A written examination shall normally last a minimum of one hour for one unit course.

3.4. External Examiners' System

The external examiner system is adopted and it is used only in the final year of the undergraduate programme to assess final year courses and projects, and to certify the overall performance of the graduating students, as well as the quality of facilities and teaching. However, the existing practice of using different External Examiners for major subject areas in professional programmes will be continued.

3.5. Grade Point Average and Cumulative Grade Point Average

Grading of courses is done by a combination of percentage marks and letter grades translated into a graduated system of Grade Point Equivalents (GPE). For the purpose of determining a student's standing at the end of every semester, the Grade Point Average (GPA) system is used. The GPA is computed by dividing the total number of credit points (TCP) by the total number of units (TNU) for all the courses taken in the semester. The credit point for a course is computed by multiplying the number of units for the course by the GPE of the marks scored in the course. Each course shall be graded out of normalized maximum of 100 marks and assigned appropriate GPE. The GPE is illustrated bellow.

Table 3: Score and Grade Point (GP) for GPA/CGPA Calculation

(i) Credit Unit(s)	(ii) Percentage Scores	(iii) Letter Grades	(iv) Grade point (GP)	(v) Grade point Average (GPA)	(vi) Cumulative Grade Point Average (GPA)
Vary according to contact hours assigned to each course per week per semester, and according to load carried by students	70-100	A	5	Derived by multiplying (i) by (iv) & dividing the sum by the Total Credit Unit	4.50 – 5.00
	60-69	B	4		3.50 – 4.49
	50-59	C	3		2.40 – 3.49
	45-49	D	2		1.50 – 2.39
	40-44	E	1		1.00 – 1.49
	0-39	F	0		<1.00

3.6. Student Workload

A full-time student of the University shall be required to register for a minimum of fifteen (15) credit units and a maximum of twenty-four (24) credit units per semester.

3.7. Requirements for Graduation

The University requires its undergraduate students to complete and pass all courses specified and offered, including industrial attachment, where applicable, for the chosen programme of study by the Faculty/Department and approved by the Senate before graduation. Other undergraduate graduation requirements are:

- The University shall award its degree on the authority of the Senate only to students who have been found worthy in character and in learning;
- Students shall successfully complete and pass all prescribed examinations for courses required for a degree programme;
- Students shall not be involved in gross misconduct, such as, but not limited to: examination malpractice; convicted felony; other convicted criminal offences; and association with or membership of secret cult or of any organization prescribed by the university or government;

d) Students shall have earned:

- not less than 120 credit units for programmes that have durations of four years;
- not less than 150 credit units for programmes that have a duration of five years.
- The cumulative attendance should not be less than 70%.

3.8. Classification of Degrees

The classification of the first degree programmes of the university shall be determined by the Cumulative Grade Point Average (CGPA) earned by students. Professional degrees may however be awarded unclassified degrees in some disciplines. The classification shall be as follows:

Table 4: Degree Classification based on CGPA

Class of Degree	CGPA
1st Class Honours	4.50 – 5.00
2nd Class Honours (Upper Division)	3.50 – 4.49
2nd Class Honours (Lower Division)	2.50 – 3.49
3rd Class Honours	1.50 – 2.49
Pass	1.00 – 1.49
Fail	< 1.0

4. WITHDRAWAL AND PROBATION

4.1. Probation

A student whose CGPA is below 1.0 at the end of a particular academic year of study, earns a period of probation for one academic session. A student on probation is allowed to register for courses at the next higher level in addition to his/her probation level courses provided that the total number of credit units carried during that semester does not exceed 24 and

- a) The regulation in respect of student work-load is complied with

- b) The pre-requisite courses for the higher level courses have been passed

4.2. Withdrawal of Students

Conditions for voluntary withdrawal from the university shall be as follows:

- a) A student who wishes to withdraw from the university shall be required to notify the Registrar in writing through the Dean of the Faculty and the Head of Department and shall be approved by the Senate;
- b) Fresh men and women shall give notice of withdrawal not later than two weeks after matriculation. Other students shall give notice of withdrawal not later than four weeks after the beginning of the semester;
- c) A student withdrawing from the university shall be required to give the reason(s) for and the effective date of the withdrawal;
- d) A student shall retain grades earned for the semester examinations preceding the date of voluntary withdrawal;
- e) A student who voluntarily withdraws from the University shall, in order to be re-admitted, send a formal application to and receive clearance from the Registrar;
- f) Senate shall provide guidelines to be followed before a student who voluntarily withdrew can resume the programme of study after a period of withdrawal.

4.2.1. Withdrawal without Authorization

A student who withdraws from the university without the approval of the Senate shall only be considered for readmission after the case has been addressed and the approval of Senate obtained for re-admission.

4.2.2. Withdrawal for Health Reasons

A student shall be asked to withdraw for Health reasons certified by the university's Director of Health Services. Such a student may be readmitted only after a valid medical report from an approved medical doctor and certified by the Director of Health Services had been received to the effect that the student is medically fit to continue the programme.

4.2.3. Disciplinary Withdrawal and Expulsion

A student suspended on disciplinary grounds shall only be readmitted with the approval of the Senate and the Vice-Chancellor. A student shall be expelled from the University as a result of gross misconduct, examination malpractice or any other offence as determined by the laws and regulations of the University. Such a student shall be given the opportunity to defend him/herself, but shall not be readmitted to the university at any other time.

4.3. Transfer

Universities are enjoined to run comparable syllabi to enable students who transfer from one university to another transfer their credits wholly.

A student, who transfer from other universities shall be credited with only those courses, deemed relevant to the programmes, which they have already passed prior to their transfer. Such students shall however be required to pass the minimum number of units specified for graduation for the number of sessions he/she has spent in the faculty; provided that no student shall spend less than two sessions (4 semesters) in order to earn a degree. Students who transfer for any approved reason shall be credited with those units passed that are within the curriculum. Appropriate decisions on transfer cases shall be subjected to the approval of Senate on the recommendation of the faculty.

5. SIWES RULES AND REGULATIONS

5.1. Introduction

The authorities of the National Universities Commission (NUC) approved of the Students Industrial Work Experience Scheme (SIWES), for Nigerian Universities and other institutions of higher learning; for its students to undergo various training in their respective fields of studies. This scheme was brought about by the inadequate practical experience gained by the students in higher institutions after graduation, due to various factors such as: lack of relevant equipment and machineries in their institutions, etc. It is therefore mandatory for all students studying

relevant courses in Nigerian higher institutions to undergo the SIWES programme at their assigned period (level) and through the stipulated duration.

5.2. Definitions

SIWES – Students Industrial Work Experience Scheme is a programme aimed at preparing the Software Engineering student for the real industrial work situations they will meet after graduation: so they will properly fit into it. It helps bridge the gap between theory and practice by providing students with an opportunity to apply their knowledge in real work situations. It is carried out in industries outside the school.

5.3. Duration

There are two modules of the training scheme. The first module is undertaken at the end of the 200-Level of study while the second module is undertaken at the end of the 300-Level of study. They each have a duration of 3 months.

5.4. Credit Unit and Graduation Requirement

In NUN, each of the two modules of SIWES is assigned three credit units. They are required for graduation; failure in either means the student is not eligible to graduate. All required form(s) must be filled and returned at stipulated time.

5.5. SIWES Unit

SIWES is to be coordinated by the SIWES Unit of the university. CS department has a representative working under the SIWES coordinator. The SIWES coordinator and CS representative will be announced to the students.

5.6. Participation

It is mandatory for students to participate in all the activities in both modules of SIWES: orientations lectures, practical activities, log booking, report writing, etc. At least 80% attendance is required.

For SIWES the grading is done as follows:

- | | |
|---|------|
| • Employer's rating | 20% |
| • Institution's supervisor rating | 10% |
| • SIWES report including Log booking | 50% |
| • Oral interview or seminar by students | 20%. |

5.7. Forms (SIWES)

Fill and return the required forms at stipulated time.

6. STUDENTS DISCIPLINARY REGULATIONS

6.1. Aim

This document regulates the process of student conduct and discipline for the students of all levels at Nile University of Nigeria (NUN) in parallel with the regulations ratified by the Nigerian Government and the NUN Law.

6.2. Content

The regulations cover the disciplinary breaches, sanctions, disciplinary committee(s), investigation process, contesting the sanctions (appeal), and the application of the sanctions for students at all levels at NUN.

6.3. Coverage

This document covers the disciplinary issues related to the educational process during a regular day. Other issues will be addressed by the related authority.

6.4. Definitions

Board:	Disciplinary Board
Committee:	Student Welfare & Disciplinary Committee
Authority:	Disciplinary authority
Improper Material:	Any kind of materials that is considered to be unethical by the university administration.

6.5. Examination Misconduct and Disciplinary Measures

1. For any type of action that requires a disciplinary measure to be taken, the related committee that eventually handles the case will look at several points before reaching a decision. Below are some of the misconducts considered major and the utmost disciplinary measures possible for committing them.
2. The disciplinary authority will decide on the final sanction according to the nature and the degree of the breach.

Table 5: Examination Misconduct Matrix

	Breach	Reprimand & Warning	Denunciation	Rustication (1 week – 1 mon.)	Rustication (1 mon. – 1 sem.)	Rustication (1 sem. – 2 sem.)	Expulsion
1.	BAD BEHAVIOUR						
a)	Acting in a way that is not compatible with being a typical student: anti-social, immoral, unethical and indecent behaviour within the campus or in any kind of programme organized by the University outside the campus		✓	✓			
b)	Shouting or singing in a loud voice, playing / listening to a musical instrument in the buildings / campus, littering around.	✓					
c)	Using abusive language and swear words.	✓					
d)	Chewing gum, talking out loud, talking on the phone at scientific meetings		✓				
e)	Carrying out political activities on campus			✓	✓		
f)	Distributing any illegal / improper handbills; posting posters, banners, etc.		✓		✓		
g)	Religious fanaticism and intolerance.			✓	✓		
h)	Carrying arms/weapons.						✓
i)	Involvement in cultism.			✓			
j)	Communicating with outside agencies, governments or organizations for the purpose of soliciting financial sponsorship / aid that may jeopardize the interest of the university.						
k)	Preventing staff from carrying out disciplinary investigations.		✓	✓			
l)	Being sentenced with any crime against the government or because of an infamous crime.			✓	✓		
m)	Rape, transgender relationships.						
2.	DISRUPTION						
a)	Causing distraction or disturbance in lectures, meetings, and / or practicals. Obstructing and disrupting the teaching process, the right of teaching and learning directly or indirectly.		✓				
b)	Provoking people to hinder the education process.	✓					
c)	Disrupting the smooth-going of education by activities such as illegal meetings, boycotts, etc.			✓			
3.	VANDALISM / AGGRESSIVENESS						
a)	Taking away, tearing, dirtying, or writing on any formal announcements.		✓	✓			
b)	Sticking posters on places not designated for this purpose.		✓				
c)	Keeping and / or spreading improper materials banned by the administration.		✓	✓			
d)	False claims, misinforming the administration when required to provide relevant information; insubordination evidenced by refusal to respond to requests by authority.	✓	✓				
e)	Writing, drawing, carving on the university property.	✓	✓				
f)	Threatening university staff and students, cursing, slandering their names, insulting and/or attacking them.				✓	✓	
g)	Taking university staff or students out of campus by force, hindering them from doing their duties, or provoking students towards these actions					✓	

h)	Assault, physically attacking university staff and students; physical abuse, verbal abuse, threats, intimidation, harassment and coercion.				✓	✓	
i)	Talking unfavourably about the personality of the university administrators alone or as a group, producing publications, provoking students for this kind of activities.				✓	✓	
j)	Trying to enter any restricted area or areas that the student(s) is not allowed to enter for any reason, causing damage at any level.			✓	✓		
4.	IMPROPER DRESSING						
a)	Indecent dressing (please refer to the related article)	✓	✓				
b)	Dress Code***	✓	✓	✓			
5.	BAD HABITS						
a)	Plagiarism.		✓	✓			
b)	Smoking on campus.			✓	✓		
c)	Drinking alcohol on campus.				✓		✓
d)	Participating in any university activity illegal (such as lessons, meetings, seminars, etc.) drunk / having taken alcohol or narcotics.					✓	✓
e)	Gambling or organizing gambling.		✓	✓			
f)	Stealing.			✓	✓	✓	✓
g)	Using, carrying, selling, trafficking, or keeping narcotics of any kind.					✓	✓
h)	Selling or trafficking cigarettes, and/or alcohol.						✓
i)	Indecent use of internet for pornography, cyber-crime, terrorism, hacking etc.				✓	✓	✓
6.	FORGERY						
a)	Letting someone else use his/her official NUN documents, or using someone else's university documents at NUN.				✓	✓	✓
b)	Forgery in general.		✓	✓	✓	✓	✓
7.	ILLEGAL ORGANIZATIONS						
a)	Organizing meetings boycotts, riots, and obstructions in order to defame the university; causing his/her family to get involved in the activities.					✓	✓
b)	Causing discriminations and polarization in terms of language, nationality, colour, religion, and sect.					✓	✓
c)	Organizing / participating in meetings without university's consent; joining in meetings impersonating the student union representative.		✓	✓	✓	✓	✓
d)	Keeping any illegal, political / ideological banners, posters, handbills at the university.				✓	✓	✓
e)	Setting up illegal associations.		✓		✓		
f)	Securing membership with illegal organizations, contributing to their activities, and acting on behalf of them.			✓	✓		
8.	EXAMINATION MALPRACTICE						
a)	Rudeness to invigilator.		✓*	✓*			
b)	attempting to cheat		✓*	✓*			
c)	Cheating, organizing cheating				✓*	✓*	✓*
d)	Leaving exam hall without permission.	✓*	✓*	✓*			
e)	Speaking to other candidates.	✓*	✓*	✓*			
f)	Exchanging oral/written communication.	✓*	✓*	✓*			
g)	Possession of unauthorized materials.	✓*	✓*	✓*			
h)	Using inappropriate answer sheet.	✓*					

i)	Disruption of the examination.	✓*	✓*	✓*	✓*		
j)	Threatening invigilators and examiners.			✓*	✓*	✓*	✓*
k)	Use of telephone and other electronic materials in the exam hall for cheating purposes.				✓*	✓*	✓*
l)	Refusal to submit answer booklet.		✓*	✓*			
m)	Tearing papers off the examination / answer booklet.		✓*	✓*			
n)	Impersonation.				✓*	✓*	✓*

Explanations: *: Failure;

Dress Code***;

- All transparent, tight fitting, bare back and clinging dresses;
- All short dresses above knee length, including knickers and shirts with slit above the knee;
- All provocative dresses that reveal the bust, belly, waist, chest and backside;
- Dresses bearing obscene and unholy inscriptions;
- Wearing of dark glasses except on medical grounds inside the class;
- All dresses that reveal political issues

Reprimand and Warning: The student is verbally warned about the kind of action that is considered to deserve a disciplinary action. A note is taken in the student file.

Denunciation: When this is the chosen course of action as deemed by the appropriate authorities, a letter of warning is issued to the offending student with copies being filed in the student affairs.

Restitution Order: When there is a damage or loss of university property, a student may be ordered to pay for the item in question for the cost of repairing the item, or a specified outright fine amount. The bursar will be instructed to debit the student's account accordingly.

Rustication (or suspension): This refers to exemption of a student from entering/ utilizing the university's facilities, or benefiting from the school's academic or administrative bodies/structures for a stated period of time.

Expulsion (or dismissal): This action demands that a student be permanently physically removed from the university premises, and entirely exempt from all privileges of being a student at the Nile University of Nigeria on authority of the Vice Chancellor.

6.6. Disciplinary Investigation

The authorities can carry out the investigation themselves, or appoint an investigator(s) to do the task.

6.6.1. The Committee

The Dean for the students in his/her faculty as may be appointed by the Committee;
'Director of CELL' (Centre for Lifelong Learning) for students receiving education at CELL;
and Dean of Student Affair.

6.6.2. Length of Investigation

The investigation starts one day later than the day the disciplinary breach was officially brought to the attention of the disciplinary authority. In case an investigator is appointed, he/she is provided with all the details of the event and any evidences. The investigation must be completed within 15 days. If the investigator cannot complete the task with the period specified, he/she can ask for additional time with an explanatory letter. The authority that started the process can approve the request or not.

6.6.3. Investigation Method

- a. The student must submit their defence letter to the committee within the date announced on the notice board.
- b. The investigator checks the evidence, listens to witnesses, can see the side of the event, and apply to a legal expert as necessary.
- c. A report is produced for each investigation process. The report indicates the time, venue, people involved, Q & A if any, and any other relevant information related to the process. The report must be duly signed by the investigator and the people involved in that particular process.
- d. The university staffs are expected to provide any information or relevant document required by the investigator and contribute to the process.

6.6.4. Investigation Report

- a. A final report is prepared at the end of the investigation. The report includes the investigation approval, starting date of the investigation, identity of the person under investigation, disciplinary breaches, investigation steps, evidences, and the defence.
- b. The report analyses each breach and discusses if the accusations are good for taking into consideration.
- c. The report should also include a suggestive sanction (if any).
- d. Related documentation is attached to the report in a folder.
- e. The report and the folder are submitted back to the authority that opened the investigation.

6.6.5. Measures Taken During Investigation

Upon deciding that it is necessary, the investigator can require that the student(s) under investigation should be deprived of entering university buildings. The authority makes the final decision on the case.

That fact that the student might have left or withdrawn from the university, got transferred to another programme or university after he/she violated the disciplinary code shall not prevent the investigation to be opened, or finalized.

6.6.6. Overlapping of Investigations

The fact that another investigation has been started by the local authorities or the state in some other cases does not mean the investigation initiated by the university should be stopped.

In the same fashion, the fact that the student might be sentenced for the breach he/she committed will not prevent the application of the sanction by NUN.

6.6.7. Finalizing the Investigation

After thorough examination of the report and the folder by the related disciplinary authorities, the documentation is immediately delivered to the authority's disciplinary committee. The committee gets together within 5 business days and starts discussions. The committee or the disciplinary authority may ask for clarification or completion of any missing step in investigation from the investigator or a committee member as necessary.

6.7. Authorized Persons and Committees that can Impose Sanctions

The faculty board is at the same time the Disciplinary Committee of the Faculty. The Superior Disciplinary Committee is the 'of Student Welfare & Disciplinary Committee' as contracted as 'Committee' under the Senate.

The faculty board can sanction the punishments of reprimand, warning, denunciation, and Rustication (Suspension) from 1 week up to 1 month directly upon the suggestion of the dean. The Committee sanctions the punishments of Rustication (Suspension)' from 1 month upwards to 1 semester, and Expulsion upon the suggestion of the related faculty board. The faculty board is required to notify the Committee for any disciplinary sanction applied.

6.8. Disciplinary Board Meeting

The Board gathers at the time, venue, and on the date specified by the Chairman of the Board.

6.9. Regularization of the Disciplinary Board's Work

The Board's Chairman organizes the preparation of the agenda for the meeting, notification of the meeting to the board members, and smooth-running of the board's work.

6.10. Minimum Number of Members Required

The minimum number of members for the meeting to reach a decision is one more than half the total number of members of the Board

6.11. Meeting Procedure

After listening to the reporter, discussions start. The Board can listen to the investigator(s) as necessary. When it is lucid that enough has been spoken and the issue is clear, voting begins, and the result is announced by the Chairman.

6.12. Voting

1. Every member of the Board has to vote for or against the case.
2. A member cannot remain neutral.
3. It is adequate to reach a decision with the votes of the majority of the Board members.
4. In case the number of the votes that are for or against the case is equal, the majority is considered to be the Chairman's side.
5. The decision summary is translated into minutes signed by the members.
6. The authorized disciplinary board is free to accept or reject the sanction offered in the investigation report.
7. Provided that the evidence and reasons are given, the board can reach a decision for or against the student.
8. Any disciplinary breach that has not been mentioned here will be punished in parallel with the closest breach mentioned above in terms of the feature and influence of the committed action.
9. In the same manner, the students have to obey the rules of various units at the university, such as the library, the dining hall, etc.
10. Violating specific rules and regulations that are not mentioned here but announced by the related units through the university administration will require measures to be taken.
11. Important Points While Choosing a Sanction; While working on the decision, the authorized investigator and the disciplinary board will take the following into consideration:
 - a) The weight / degree of the action in question as an intensifying or lessening factor.
 - b) The intention of the student.
 - c) Whether he/she has breached the disciplinary code before.
 - d) His/her behaviour, attitude, and actions.
 - e) Whether he/she has been remorseful.
 - f) If the disciplinary rule(s) has been breached together with a person/people from outside the university within the campus area, an upper degree disciplinary measure is taken.

- g) If the disciplinary offence has been committed by a group of students and the group members could not have been identified one by one, the group will receive a punishment as a whole.

6.13. Application and Contesting Decisions

6.13.1. Communication of the Sanctions

To communicate the outcome of the investigation, the authority that initiated the investigation will do the following:

1. Students who have been imposed a sanction of suspension or an above degree sanction will be exempt from any social contributions or scholarships.
2. When there is a damage or loss of university property, a student may be ordered to pay for the item in question for the cost of repairing the item, or a specified outright fine amount. The bursar will be instructed to debit the student's account accordingly.
3. If a student violates the same / similar rule more than once during his/her period of study, he/she will be punished with a sanction that is one degree heavier.
4. Examination malpractice: The decisions will be put into student's file.
5. Disciplinary sanctions can be announced at all the units on campus
6. Notify the concerned student;
7. University may notify the mother, father, and in case both parents are deceased, the legal guardian.
8. University may notify the scholarship providers, whether governmental, private, or person;
9. If the sanction is 'expulsion', in addition to the above mentioned: Security forces (if the breach requires action at the official level).

6.13.2. Application

Disciplinary sanctions will be put into practice on the day they are announced unless stated otherwise in the minutes of the board meeting authorized.

6.13.3. Contesting the Decisions and Duration

Student cannot contest the sanctions of reprimand, warning, denunciation, restitution order, and suspension from 1 week to 1 month. Students can contest the decisions of suspension from 1 semester to 2 semesters, and expulsion to the Committee through the HOD, the Dean, and the VC within 3 days after the decision has been announced. The Committee thoroughly analyses the case and either rejects the application or accepts it.

6.14. MISCELLANEOUS

6.14.1. Notification and Notification to Address

All kinds of notifications will be made to the address the student has submitted upon registration or are posted as a notification at the university. The students carry the sole responsibility for the accuracy of the current address the university holds in the files. Therefore, the students cannot claim that the university has not notified them in case the address was wrong.

6.14.2. Archiving the Related Files

The Disciplinary Authority keeps the related files. The disciplinary sanction is noted down to the student file.

6.14.3. Method of Correspondence

Regular postal service and the student email are utilized for communicating with persons. If a document is handed to a person, a 'received' slip is kept in the file.

6.14.4. Prohibitions

Confidentiality prevails at all steps of the investigation. Any Staff of the institution who violates secrecy of any procedure will be treated according to the Staff Disciplinary Regulations.

7. TOPICS PRE-EXAMINATION SHOULD BE KNOWN

The objective of the written examination is to assess how much the students know about the courses which lecturers have taught them for particular semester in accordance with the course

description and syllabus. Therefore, the examination procedures that lecturers should adopt in the conduct of examinations include the following:

7.1. Before the Examination Week

- The examination timetable is posted to the related notice board or University website beforehand.
- The Examination Committee will announce the seating order at the notice boards (if arranged for certain examinations).
- Students who have clashes in the timetables are strongly advised to contact the Examination Committee Secretary immediately as they have the sole responsibility for the issue.
- Students with less than 70% attendance, disciplinary, and/or financial issues will not be allowed to sit the examinations.
- Students with no NUN ID Cards will not be admitted to the examinations. Students must carry their Students ID card at all times, particularly during the examination period.
- End of semester examinations are held at the end of each Semester. Examinations are conducted for courses taught in the department. The specific regulations are:
- Any student who is absent from any examination without the University's approval and has not withdrawn from the course of study shall be graded 'F' for such courses.
- A candidate shall not be allowed to enter the examination hall 30 minutes after the commencement of examination.
- A candidate shall not be allowed to leave the examination hall within 30 minutes after the commencement of examination except under exceptional circumstances approved by the Chief invigilator. No candidate shall leave the examination hall during the last 15 minutes of the examination.
- Candidates shall comply with all instructions on allotted time for commencement and stoppage of the examination.
- An irregularity in the conduct of the examination including any breach of the above regulations by any Student, such a person shall be arraigned before the Faculty and the University for Disciplinary Action.

7.2. Prepare a study guide:

Before the examination, a lecturer should conduct a revision for the students necessary to prepare them for the examination. Likewise, a study guide should be provided for the students.

7.3. Just Before the Examination Starts

- The candidates have the responsibility of removing implicating materials found around the examination hall including cell phones, computers of any kind, papers, bags, books, handouts, and any foreign materials.
- Candidates are required to provide for themselves any necessary examination materials as no exchange of lending or borrowing of any item is permissible during examination.
- Students are required to demonstrate their valid University ID Cards as the invigilators will be checking their identities.
- The students are advised to arrive at the examination hall 15 minutes prior to the start time to enable them properly identify themselves and take the proper sitting order.
- Students are normally not allowed to take breaks during the examination. Students are advised to contact the Chief Invigilator if there is a only current medical issue to obtain permission for taking a short break during the exam.

The lecturers are required to explain the format of the examination and answer the questions coming from the students.

The written examination will normally be on the basis of not less than one hour, and not more than one-half hours for each ten hours of lecture. In any case, the time allowed for any written paper, will not exceed three hours.

7.4. During the Examination

- A candidate shall not be allowed to enter the examination hall 30 minutes after the commencement of examination with no extra time allowed.
- A candidate shall not be allowed to leave the examination hall within 30 minutes after the commencement of examination except under exceptional circumstances approved by the Chief invigilator. No candidate shall leave the examination hall during the last 15 minutes of the examination.
- Students are advised to:
 1. Carry their ID Cards at all times.
 2. Proofread their answers if there is still time left.
 3. Not to fold the answer booklet.
- Disciplinary actions:
 1. Silence is expected from the students at all times. Any distraction or disturbance in the exam room will not be tolerated.
 2. Students, however, will review the questions at the beginning of the examination and communicate with the invigilator for technical reasons.
 3. Candidates shall comply with all the instructions on allotted time for commencement and stoppage of the examination.
 4. Any student who attempts to manipulate internal or external examiners in any fashion will face disciplinary action.
 5. Any breach of the examination regulations by any student will be brought to the Examination Committee for action.

Please be alert the time: Invigilators should make sure that they can see all the students or the group of students under their responsibility all the time. This means invigilators cannot read newspapers, do their own work, turn their back to the students, etc.

Do not:

- Let the students change their seats during examination;
- Let the students exchange stationary;

- Provide a student with an extra booklet unless he/she has totally utilized the first one. The successive answer booklet(s) will be provided with their covers off, and the first page signed by the invigilators;
- Let the students who finished and went out talk loudly in the corridors

7.4.1. Fill out the form:

Also 10 minutes into the exam, the invigilators should distribute the attendance form and ensure that only the students present duly sign the form.

7.4.2. Announcement of results:

The final results will be uploaded to the automation system after due approval by the Senate.

7.5. Contesting Examination Results

- A candidate wishing to contest his / her examination results may forward a letter to the Head of Department including the full details of the examination, that is, course number and title, semester, and the reason of the request in 10 business days after the results are announced.
- The department will re-evaluate the result then in case of change of the grade the report will be submitted to the faculty board.
- In case of change of the grade The Faculty Board will present a report for the attention of the Vice Chancellor in 10 business days who in turn forward the report for the Senate's consideration.

7.6. Students' Evaluation of Courses

There should be an established avenue put in place offering opportunity to students to evaluate courses delivered to them at the end of each semester. This should be an integral component of the course credit system; serving as feedback mechanism for achieving the following:

- i) Improvement in the effectiveness of course delivery.
- ii) Continual update of lecture materials to incorporate emerging new concepts.

- iii) Effective usage of teaching aids and tools to maximize impact of knowledge on students.
- iv) Improvement in students' performance through effective delivery of tutorials, timely in presentation of continuous assessment and high-quality examination. In order to achieve effective learning, all students should normally be permitted to evaluate those courses registered at the end of each semester, preferably before the final semester examinations. It is very important that students' evaluation of courses be administered fairly and transparently through the use of well-designed questionnaires, maintain confidentiality demanded by such exercise and apply their scientifically processed outcome to improving effective course delivery in all ramifications.

7.7. Communication Channel

Every student is given an official e-mail by the school comprising of their ID No. i.e., (Student ID)@nileuniversity.edu.ng. Students should use this email for all official communications. Also the Faculty has opened a Google group and added all students to it: it is to be used for announcements.

SOFTWARE ENGINEERING CURRICULUM

100 Level (Harmattan)

Course Code	Course Title	Unit(s)	Status	LH	PH
GST111	Communication in English	2	C	15	45
NUN-GET 101	Introduction to Engineering	3	C	45	-
MTH101	Elementary Mathematics I	2	C	30	0
PHY101	General Physics I	2	C	30	0
PHY107	General Practical Physics I	1	C	0	45
STA 111	Descriptive Statistics	3	C	45	0
COS101	Introduction to Computing Sciences	3	C	30	45
NUN-COS 103	Introduction to AI and Machine Learning	2	C	30	
TOTAL		18			

100 Level (Rain)

Course Code	Course Title	Unit(s)	Status	LH	PH
GST112	Nigerian Peoples & Culture	2	C	30	0
NUN-COS 112	Introduction to Green Technology	2	C	30	-
MTH102	Elementary Mathematics II	2	C	30	0
PHY102	General Physics II	2	C	30	0
PHY108	General Practical Physics II	1	C	0	45
SEN102	Introduction to Software Engineering	2	C	30	0
COS102	Introduction to Problem Solving	3	C	30	45
GST105	IT and Library Skills	2	E	30	0
TOTAL		16			

200 Level (Harmattan)

Course Code	Course Title	Unit(s)	Status	LH	PH
NUN-SEN 210	Software Engineering Process	2	C	30	-
NUN-SEN 207	Software Requirements and Design	3	C	45	-
MTH 201	Mathematical Methods I	2	C	30	0
COS201	Computer Programming I	3	C	30	45
NUN-SEN 201	Introduction to Web Technologies	3	C	15	45
CSC203	Discrete Structures	2	C	30	0
IFT 211	Digital Logic Design	2	C	15	45
TOTAL		16			

200 Level (Rain)

Course Code	Course Title	Unit(s)	Status	LH	PH
GST 212	Philosophy, Logic and Human Existence	2	C	30	0
ENT212	Entrepreneurship and Innovation	2	C	30	0
MTH 202	Mathematical Methods II	2	C	30	0
COS202	Computer Programming II	3	C	30	45
SEN299	SIWES I	3	C	0	135
INS204	System Analysis and Design	3	C	30	45
IFT 212	Computer Architecture and Organisation	2	C	15	45
NUN-COS 204	Digital Media Communication	2	C	15	45
TOTAL		19			

300 Level (Harmattan)

Course Code	Course Title	Unit(s)	Status	LH	PH
SEN301	Software Construction	2	C	15	45
NUN-DTS 304	Database Management	3	C	30	45
NUN-ICT 306	Data Communications and Network	3	C	30	45
NUN-IFT 311	Web Application Development	3	E	30	45
NUN-SEN 331	Engineering Mobile Application	3	E	30	45
CSC301	Data Structures	3	C	15	45
NUN-COS 305	Remote Work and Virtual Collaboration	2	C	30	
TOTAL		19			

300 Level (Rain)

Course Code	Course Title	Unit(s)	Status	LH	PH
GST 312	Peace and Conflict Resolution	2	C	30	0
ENT312	Venture Creation	2	C	15	45
SEN304	Software Testing and Quality Assurance	2	C	15	45
SEN306	Object-Oriented Analysis and Design	3	C	15	45
SEN322	Software Engineering Innovation and New Technology	2	C	15	45
SEN399	SIWES II	3	C	0	135
CSC308	Operating Systems	3	C	30	45
TOTAL		17			

400 Level (Harmattan)

Course Code	Course Title	Unit(s)	Status	LH	PH
COS409	Research Methodology and Technical Report Writing	3	C	30	0
SEN401	Software Configuration Management and Maintenance	2	C	15	45
NUN-SEN 405	Design and Analysis of Algorithms	2	E	30	-
NUN-SEN 417	Compiler Construction	2	E	30	-
SEN441	Software Architecture and Design	2	C	15	45
INS 401	Project Management	2	C	30	0
SEN497	Final Year Student's Project I	3	C	0	135
TOTAL		16			

400 Level (Rain)

Course Code	Course Title	Unit(s)	Status	LH	PH
NUN-SEN 404	Software Reverse Engineering	2	C	15	45
NUN-CSC 408	Artificial Intelligent	3	C	15	45
NUN-SEN 410	Human Computer Interaction	3	C	30	45
NUN-SEN 444	Secure Code Development	3	C	45	-
NUN-SEN 407	Software Engr. Professional Practice	2	C	30	-
SEN498	Final Year Student's Project II	3	C	0	135
TOTAL		16			

8. ELECTIVE

Course Code	Course Title	Unit(s)	Status	LH	PH
SEN 411	AI and Expert Systems	2	E	2	2
SEN 412	Special Topics in SE	3	E	3	-
NUN-SEN 405	Design and Analysis of Algorithms	2	E	2	-
SEN 415	Embedded Systems	2	E	2	2
NUN-SEN 417	Compiler Construction	2	E	2	-

NUN-GET 101: Introduction to Engineering (3units; Compulsory; L=45, P=0)

Course Contents

An introduction to the engineering profession. the engineering disciplines. The engineering design process; teamwork, and problem solving. Engineer-in-Society. Philosophy of science and engineering. History of engineering. Engineering professional bodies and engineering societies. Engineers' code of conducts and ethics. Engineers and nation building, economy, politics, business. Safety in engineering and introduction to risk analysis. Introduction to Microsoft Office Productivity Tools: - Word, PowerPoint, Excel. Introduction to engineering tools - MATLAB, Mathematica, etc.;

NUN-SEN 104: Introduction to Web Technologies (2 units; Compulsory; L=15; P=45)

Course Contents

Introduction to the internet. The World Wide Web (WWW) and web development. WWW as a platform for interactive applications. Content publishing, and social services. The role of HTTP and HTTPS in the context of web applications. Roles and operations of web browsers and the webserver. Interacting with web applications through forms, and using style sheets to separate document structure and document formatting. Web development tools and frameworks. Build a simple website that: organizes information effectively, uses valid HTML and CSS, and applies appropriate web standards from standards bodies such as W3C. HTTP communication protocol. The mark-up languages XHTML, and XML, the CSS and XSLT standards for formatting and transforming web content. Interactive graphics and multimedia content on the web. Client-side programming using JavaScript. Impact of the world wide web on people's lives over time.

Lab Work: Using simple form-based web applications; developing simple websites using web development tools and frameworks; using the mark-up languages HTML, XHTML and XML; using JavaScript.

NUN-SEN 112- Green Technology

(2 units; Compulsory; LH=30, p=0)

Course Contents

Introduction to green computing. Importance of green information technologies. Environment management. Renewable energy production. Energy efficiency. Governance and regulatory issues land use. Waste management and recycling. Traditional/indigenous technology, production technologies for bio-products, environmental improvement, minimizing power usage. Business process reengineering for sustainability. Sustainable hardware. Going paperless. Technology company case studies. Data centre design and redesign. Virtualization the future; staying green.

NUN-SEN 207- Software Requirements Engineering (3 units; Compulsory; L=45, P=0)

Course Contents

Definition of requirements engineering; the importance of requirements engineering. Understanding software requirement. Types of software requirements; product and process requirements, functional and nonfunctional requirements, emergent properties, quantifiable requirements, system and software requirements. Requirement engineering processes. Requirements elicitation; activities, challenges. User involvement in requirements elicitation, requirements sources, stakeholder. Understanding the different stakeholders and their roles. Requirements elicitation techniques. Techniques of expressing requirements; user stories. Other techniques; flow chart, data flow graph, use case. Requirements modeling; modeling, process models and actors. Managing requirements elicitation. Documenting requirements; Software Requirements Specification (SRS). Management requirements; requirements change, requirements attributes, requirements tracing, measuring requirements, managing change request. Software Requirements Tools; UML diagrams. Requirements cost estimation.

NUN-SEN 210 - Software Engineering Process (2 units; Compulsory; L=45; P=0)

Course Contents

Software process definition. Software process management and infrastructure. Software life cycles. Categories of software process. Software life cycle models. Software process adaptation. Practical considerations. Software process assessment and improvement. Software process assessment models. Software process assessment methods. Software process improvement models, continuous and staged software process rating. Software measurement; software process and product measurement. Quality of measurement results. Software Information Models. Software process measurement techniques; software engineering process tools.

NUN-DTS 304 Database Management

(3 Units; Core; L = 30; P = 45)

Course contents

Information Management Concepts. Information storage & retrieval. Information management applications. Information capture and representation. Analysis and indexing - search, retrieval, information privacy. Integrity and security. Scalability, Efficiency and Effectiveness. Introduction to database systems. Components of database systems. DBMS functions. Database architecture and data independence. Database query language. Conceptual models. Relational data models. Semi-structured data models. Relational theory and languages. Database Design. Database security and integrity. Introduction to query processing and optimisation. Introduction to concurrency and recovery.

Lab work: Practical exercise on information representation, capture, storage and retrieval. Learn how to analyse data and index for easy searching and indexing. Practical on creating database files and models. How to create and use various database designs. How to query the created database. Methods of concurrency and recovery in database. Learn how to secure the database.

NUN-ICT 305 Data Communications and Network (3 Units; Core; L = 30; P = 45)**Course contents**

Types and sources of data. Simple communications network. Transmission definitions, one way transmission, half duplex transmission, transmission codes, transmission modes, parallel transmission, serial transmission, bit synchronisation, character synchronisation, character synchronisation, synchronous transmission, asynchronous transmission, efficiency of transmission. Introduction to network protocol. Seven Layer ISO-OSI standard protocols and network architecture. Transport protocols, session services protocols, and other protocols. Institute of Electrical and Electronics Engineering 802 standards. Error control and Data Compression: Forward Error Control; error detection methods; parity checking; linear block codes, cyclic redundancy checking; feedback error control, data compression, Huffman coding and dynamic Huffman coding. Local Area Networks: medium access control techniques – Ethernet, token bus and token ring; fibre distributed data interface, metropolitan area network. Peer-to-peer, Client Server. Client-Server Requirements: GUI design standards, interface independence, platform independence, transaction processing, connectivity, reliability, backup, and recovery mechanisms. Features and benefits of major recovery mechanisms. Network OS: (e.g., Novell NetWare, UNIX/LINUX, OS/2 & Windows NT). INTERNET: Definition, architecture, services, internet addressing. Internet protocol, IPv4, IPv6.

Lab Work: Demonstration of simple communications networks. Illustration of applications at the various levels of the OSI model. Demonstration of different types of Local Area Networks (LANs). Illustration of Metropolitan Area Networks. Illustration of Error Detection and Error Correction techniques. Demonstration of Network Operating Systems.

NUN-SEN 311 Web Application Development (3 Units; Elective; L = 30; P = 45)**Course contents**

Introduction to Web Application and technologies. The three-Tier Architectures; the Client Tier, the Middle Tier, the Database Tier. Introduction to framework-based web development using a contemporary language like PHP and ASP.net. Principles of web pages (dynamic and static) and website design. The tool used in web development. Client-side and server-side languages. Creation of interactive, dynamic websites using a common web architecture and object-based database access. Design, implementation, and testing of web-based applications including related

software, databases, interfaces, and digital media. Standard object models, and the use of server-side programmes for database and file access; testing, software quality assurance; and the process of publishing Web sites. Hands-on PHP and Python programme using open-source software (Apache, PHP, Python, JavaScript, and MySQL). Programming for web development includes control structures, objects, functions, and the use of composite data types. Deploying dynamic content using JavaScript. Designing and developing dynamic web pages and creating, validating, transforming, and formatting data using PHP. Validation on the Client Server. A Brief Summary of an HTTP Client/Server. Negotiation Sessions. Authentication and Security.

Lab Work: Simple PHP programming. Design of simple web pages. Creation of dynamic websites. Design of client-side and server-side programmes. Demonstration of web-based applications with database access. Use of JavaScript to develop dynamic content. Use of Python to develop dynamic web pages.

NUN-SEN 331 Engineering Mobile Application (3 Units; Elective; L = 30; P = 45)

Course contents

Overview of mobile devices. Types of mobile devices and capabilities. Mobile Operating Systems; types and Architecture. Mobile applications and their classification. Android architecture and its application. Developing Mobile Application; tools and techniques. Extending to major components. Activities. Services. Broadcast receivers. Development of interactive UI. App Widget libraries. Web-based services. Animation. Database engine. SQL. Multithreading. Mobile app testing on both virtual and real device.

Lab work

Installation and configuration of mobile IDE for Android. Practice UI design for mobile. Developing UI components for app. Creating app widgets. Creating functionalities for widgets. Integrating multiple widget component on a UI. Compiling an app. Creation of emulation device and configuration. Testing app based on emulator device. Testing on real device.

NUN-SEN 404 Software Reverse Engineering (3 Units; Elective; L = 30; P = 45)

Course contents

Introduction to reverse engineering (RE). Application of RE in software Engineering; software redocumentation, software design regeneration, software code extraction, software testing, software security issue. Uses of reverse engineering. Methodology and tools used in reverse engineering. Review of source code transformation tools; compiler and assembler. Code compilation process. Discussion on basic source code reversing tools; de-compiler, de-assembler, debugger. The tricks and pitfalls of analysing compiled code. Identifying calling conventions. Code analysis; Control Flow Analysis, Data Flow Analysis, Identifying Control of software through control flow graph and data flow graph. Generating control flow graphs. Using a debugger to aid reverse engineering. Static analysis tools and techniques for reverse engineering. Dynamic analysis tools and techniques for reverse engineering. Reversing mobile applications to extract code and documentation. Challenges in reversing software. Obfuscation; its application, importance and tools.

Lab Work: practice and study code compilation in Java using eclipse IDE. Reverse applications using generic reversing tools. Practice debugging of reverse engineered code. Demonstrate reversing of APK file to bite code. Demonstrate reversing bytecode to source. Practice code

analysis to generate control flow graph of app. Demonstrate extraction of source code after reversing.

NUN-SEN 405 Design and Analysis of Algorithms (2 Units; Elective; L = 30; P = 0)

Course contents

Review of data structures. Introduction to algorithms. Analysis of algorithms. Complexity Notations. Complexity Analysis techniques. Time complexity for sorting, searching, and complexity classes. Recurrences. Graph algorithms. Adjacency Matrix and Adjacency List. Shortest Path Algorithms; Dijkstra's algorithm, Floyd-Warshall Algorithm, Bellman-Ford Algorithm. Backtracking. Johnson Algorithm Divide-and-conquer approach. Greedy algorithm. A* algorithm. Topological sort. Minimum spanning tree. Shortest paths. Backtracking. Backtracking applications in games. String matching. Other Algorithmic Techniques; Dynamic programming. Longest common subsequence. Theory of NP-completeness. Turing machines. The halting problem.

NUN-SEN 407 Professional Practice in IT (2 Units; Core; L = 30; P = 0)

Course contents

Introduction to professional practice. Organisational and professional communication. Mindfulness. Teamwork. the nature of the IT professions. The role of professional associations. Codes of ethics. Professional conduct. Nature and role of professional societies. Software engineering standards. Employment contracts. Legal issues. Trade-off analysis. Group dynamics and psychology. Dynamics of working in teams/groups. Individual cognition dealing with problem complexity. Interacting with stakeholders. Dealing with uncertainty and ambiguity. Dealing with multicultural environments. Communication skills. Ready understanding and summarizing. Writing. Teams and group communication. Presentation skills

NUN-CSC 408 Artificial Intelligent (2 Units; Core; L = 15; P = 45)

Course contents

Overview of Artificial Intelligence. History of AI. Goals of AI. AI Technique. Types of AI. Branches and applications of AI. Advantages and Disadvantages. Introduction to Intelligent Agents. Agent Performance, Examples of Agents, Agent Faculties, Rationality, Agent Environment. Agent Architectures. Search. General Classes of AI Search Algorithm Problems. Problem Solving by Search. Types of AI Search Techniques and Strategies. Introduction to the types of problems and techniques in AI. Problem-Solving methods. Major structures used in AI programmes. Knowledge Representation. KR and Reasoning Challenges. KR Languages. Knowledge representation techniques such as predicate logic, non-monotonic logic, and probabilistic reasoning. Semantic Network - types of relationships, semantic network inheritance, types and components. Introduction to Frames. Natural Language Processing (NLP). Introduction to natural language understanding and various syntactic and semantic structures. Introduction to Expert Systems - characteristics, components, types, requirements, technology, development. Programming Languages for AI. Introduction to computer image recognition.

Lab work: Group practical in (i) Turing test practical - Students can act out their own version of the Turing test (iii) Facial recognition practical to aid in teaching students how machine learning works with students simulating a facial recognition algorithm. Practical applications of NLP in

groups – (i) Question Answering focuses on building systems that automatically answer the questions asked by humans in a natural language (ii) Spam detection application for detecting unwanted e-mails getting to a user's inbox (iii) Sentiment analysis/opinion mining should be used on the web to analyse the attitude, behaviour, and emotional state of the sender, implemented through a combination of NLP and statistics (iv) Practical exercise of machine translation used to translate text or speech from one natural language to another natural language such as the Google Translator (v) Developing a model to provide word processor software for the spelling correction (vi) Developing a model for speech recognition for converting spoken words into text (vii) Implementing a Chatbot to provide the staff/student's chat services. OR Group Practical exercise on agents and its environment using simulation of a colony of ants foraging for food; model simulating a message between agents; model simulating the flocking behaviour of birds; model to apply standard search algorithm to the classic search problem of missionaries and cannibals, and how to use communicating agents for searching networks.

NUN-SEN 417 Compiler Construction

(2 Units; Elective; L = 30; P = 0)

Course contents

Review of compilers. Assemblers. Interpreters. Structure of a compiler. Functional aspects of a typical compiler. Syntax and semantics. Lexical analysis. Expression analysis. Code generation. Functional relationship between Lexical and Expression analysis. Internal form of source program. Use of a standard compiler. FORTRAN/COBOL/PL as a working vehicle. Error detection. Error recovery. Grammars. Languages. The parsing problem.

NUN-SEN 441 Human Computer Interaction

(3 Units; Core; L = 30; P = 45)

Course contents

Introduces the principles of user interface development. How to design good user interfaces. Human capabilities to drive design techniques. Task analysis. User-centered design. Iterative design. Usability guidelines. Interaction styles. Graphic design principles. Techniques for building user interfaces. Low-fidelity prototypes. Wizard of Oz. Prototyping tools. Input models. Output models. Model-view-controller. Layout, constraints, and toolkits. Evaluation Techniques for evaluating. Measuring interface usability. Heuristic evaluation. Predictive evaluation. User testing.

Lab Work: Illustration of the principles of HCI design. Practice on GUI design and programming. Demonstration of some GUI toolkits. Practical evaluation of GUIs

NUN-SEN 444 Secure Code Development (3 Units; Core; L = 45; P = 0)**Course contents**

History and terminology. Security mindset. Common software security vulnerabilities, threats, and attack surfaces. Anti-patterns in software security. Secure coding principles and practice. Secure coding standards and conventions. Strategies for secure coding; use of patterns and frameworks in software design and architecture, defensive programming, input validation, data sanitization, exception handling, security by design. Software lifecycle secure coding approach. Secure coding practice in design and architecture, implementation, testing, operations.

Information assurance analysis model. Disaster recovery. Forensics. Authentication. Intrusion detection. Operational issues-trends. Auditing. Standards enforcement. Legal issues. Policy-creation of policies. Maintenance of policies. Prevention and avoidance. Incident response (forensics). Domain integration (physical, network, internet, etc.). Social engineering. Denial of service. Protocol attacks. Active and passive attacks. Buffer overflow attacks. Malware. Security domains. Security awareness and possible domains. Forensics legal systems. Digital forensics. Relationship of other forensic disciplines. Rules of evidence. Search and seizure. Digital evidence. Media analysis. Security services. Threat analysis model. Vulnerabilities.

10. SOFTWARE ENGINEERING LABORATORIES

The Software Engineering department operates two (2) laboratories as well as a network laboratory for use of the students. The laboratories are equipped with high-speed high-performance computers as well as projectors. The computer systems are fully loaded with relevant software to support the program course structure.

1. List of Laboratories

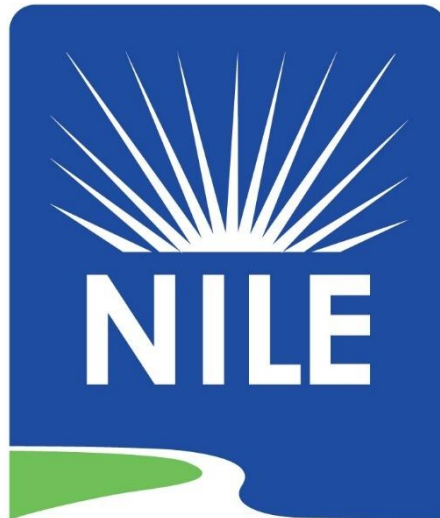
- Computer Laboratory I – Room 130
- Computer Laboratory II – Room 132
- Network Laboratory – Room 144

2. Laboratory Rules and Policies

The Information Technology Services Computer laboratories are for the exclusive use of students, faculty and staff. The following rules will help us ensure that you get the most out of the resources located within these facilities.

- Alteration of the equipment or furniture is prohibited.
- No food or drink is permitted inside this facility. We have provided a designated area to leave any food or drink you may be carrying.
- Game playing and use of the lab for business or other non-scholarly activities is prohibited.

NUN is an equal opportunity institution. It does not discriminate against any member of its community on the basis of sex, race, nationality, ancestry, creed, marital or parental status, or physical, mental, emotional, or learning disabilities in its educational programmes and activities.



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